Client Capabilities

Cenomi Chatbot — Capabilities & Roadmap

Product: Cenomi AI Mall Concierge
Version: v1.6
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Status: Production-Ready

Overview

The Cenomi AI Mall Concierge is a conversational AI assistant embedded at the point of visit — on digital kiosks, web portals, and mobile touchpoints — that helps mall visitors find what they need, plan their visit, and discover experiences they might otherwise miss. It speaks the language of the visitor, understands context, and responds like a knowledgeable human concierge would.

Powered by a Large Language Model (LLM) pipeline built on a directed graph architecture, the concierge adapts its behaviour to each query: it looks up exact facts when precision matters, offers curated recommendations when context is richer, and builds itineraries when a visitor needs to plan their whole visit.

What the Chatbot Can Do Today

1. Factual Lookups — Instant, Precise Answers

The chatbot has complete factual knowledge of each configured mall and answers direct questions immediately.

Capability	Example Queries
Store locations	"Where is Zara?", "How do I get to Swarovski?"
Operating hours	"What time does the mall close?", "Is it open on Fridays?"
Cinema listings	"What movies are showing?", "Any kids' movies today?", "What time is the first show?"
Services & facilities	"Where is the prayer room?", "Is there an ATM?", "Do you have wheelchairs?"
Parking	"Is parking free?", "Where can I park near Gate 3?"

Capability	Example Queries
Brand availability	“Is Nike here?”, “Do you have H&M?”, “Is there a Starbucks?”
Cross-mall brand search	“Which Cenomi malls have Zara?”, “Where can I find a cinema nearby?”
Mall overview	“Tell me about this mall”, “What’s here?”, “Is this mall family-friendly?”

Accuracy: Responses draw directly from the mall’s structured data store — store units, floor numbers, zone names, and directions are precise.

2. Guided Recommendations — Personalised Suggestions

When a visitor has an intent but needs help narrowing it down, the chatbot acts as a curator — filtering, ranking, and presenting the best-fit options for their context.

Capability	Example Queries
Dining recommendations	“Where should we eat?”, “Something nice for a date”, “Quick food before the movie”
Shopping by category	“Where can I find shoes?”, “I need a jacket”, “Looking for perfume as a gift”
Budget filtering	“Something affordable”, “My budget is 300 SAR”, “Not too expensive”
Audience-aware suggestions	“For my 5-year-old son”, “She likes bags”, “Trendy options for teens”
Occasion-driven picks	“I’m a bridesmaid”, “It’s our anniversary”, “Shopping for back to school”
Constraint refinement	“Actually, she’s vegan”, “Something that isn’t too loud”, “More casual”
Correction handling	“Actually I’m alone, no kids”, “I meant for a girl, not a boy”

The chatbot **remembers prior turns** in a session — so “something more affordable” after a shopping recommendation correctly applies the budget filter to the prior suggestion, without the visitor re-explaining.

3. Multi-Domain Planning — Full Visit Itineraries

For visitors planning a complete outing, the chatbot composes structured plans that span multiple activities, zones, and timings.

Capability	Example Queries
Movie + meal plans	“Food and movies”, “We want to catch a movie and then eat”
Timed visit plans	“I have 2 hours — what should I prioritise?”, “Map out a 4-hour family visit”
Pre / post movie routing	“Something quick before the movie starts”, “Dessert after the film”
Occasion itineraries	“Can you give me a full itinerary for tonight?”, “Plan our anniversary evening”
Group outings	“Team outing for 10 people — food and activities”, “Fun for 5 teens on a budget”
Shopping day plans	“I need a 2.5-hour back-to-school shopping route”
Bridal party planning	“Map out a full bridal party day”

Plans are structured, sequenced, and anchored to real zones and stores — not generic advice.

4. Context Awareness — Understands Who You Are

The chatbot continuously builds a scene model of the visitor’s situation and uses it to tailor every subsequent response.

Context Signal	How It’s Used
Companions (family, kids, group, couple)	Adjusts recommendations for appropriate age groups, dining style, activity type
Occasion (anniversary, birthday, wedding, team outing)	Selects the right tone, venue suggestions, and experience anchors
Budget declared (“around 300 SAR”)	Filters all subsequent recommendations to budget-appropriate options
Target person (“it’s for my 7-year-old daughter”)	Scopes suggestions to the intended recipient
Excluded domains (“no food”, “skip dining”)	Removes that domain from all further suggestions
Prior refinements	Preserved across turns so the visitor never has to repeat themselves

5. Smalltalk & Conversational Handling

The chatbot handles the full range of conversational turns a real concierge would encounter.

Category	Examples
Greetings	“Hi”, “Hello”, “Hey” — receives a warm, context-setting welcome
Gratitude	“Thank you”, “Shukran” — acknowledged naturally with a soft next-step offer
Farewells	“Goodbye”, “See you” — warm close with invitation to return
Identity questions	“Who are you?”, “Are you a real person?”, “Are you a bot?” — honest, reassuring response
Emotional turns	“I’m so bored”, “I feel overwhelmed” — empathetic acknowledgement, gentle redirection
Crisis language	Detected and handled with care — no clinical flags, no panic, calm supportive response
Off-topic queries	“What’s the weather?”, “Can you book a taxi?” — gracefully declined with a redirect to what the chatbot can help with
Garbage input	“asdf”, “a” — recovers gracefully and offers to help

6. Multi-Mall Support & Cross-Mall Search

The chatbot is configured for **5 Cenomi mall properties** and supports both single-mall conversations and cross-mall brand queries.

Mall	City	Stores	Dining	Cinema
Al Nakheel Plaza (Buraidah)	Buraidah	83	10	✅ Muvi
Al Nakheel Plaza (Riyadh)	Riyadh	83	10	✅ Muvi
Al Ahsa Mall	Al Ahsa	85	12	✅ Muvi
The View Mall	Riyadh	125	28	✅ Muvi
Al Nakheel Mall	Riyadh	219	37	✅ Muvi
Mall of Arabia	Riyadh	242	44	✅ Muvi

Cross-mall queries (“Which Cenomi malls have Starbucks?”) are answered with a ranked list spanning all configured properties.

7. Observability & Feedback

Each conversation generates rich observability data used to improve the system over time.

- **Debug payload** on every response: response mode, confidence level, active playbook, flow type, retrieval status, latency per node
- **Implicit feedback detection:** Detects signals like topic switches, refinements, and repeated requests to identify satisfaction signals
- **Feedback normalisation:** Collected feedback is normalised for quality analysis
- **Session memory:** Full conversation history preserved within a session; the concierge never loses track of what was said

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How It Works (Technical Summary for Stakeholders)

The chatbot uses a **12-node LLM-first pipeline** built on LangGraph:

1. **Interpret Turn** — LLM classifies the visitor's intent, extracts scene context (companions, occasion, budget), and selects a response strategy
2. **Route Flow** — Decides between the factual path (exact data lookup) and the concierge path (recommendation/planning)
3. **Resolve Playbooks** — Matches the intent to pre-defined experience playbooks (28 per mall) that guide the response shape
4. **Compose Context** — Assembles the relevant stores, dining options, services, and facts into a prioritised context window
5. **Rank & Dedupe** — Scores and orders entities by relevance to the visitor's stated intent, companions, and constraints
6. **Generate Response** — LLM generates a fluent, concierge-quality reply grounded in the composed context
7. **Update Scene Memory** — Captures new context signals (new companions, excluded domains) for use in subsequent turns

LLM used: GPT-4o-mini (OpenAI) — balances response quality with cost and speed
Average response time: 7.5 seconds end-to-end
Data freshness: Mall data updated through a structured ingest pipeline; movies, offers, and events reflect the canonical data snapshot

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Playbook Library (28 Scenarios per Mall)

Each mall is configured with 28 response playbooks — pre-defined scenario templates that govern the recommendation strategy, entity selection, and response shape for common visitor journeys:

Category	Playbooks
Family visits	Family visit plan, family shopping with child, child activity while parents shop, kids entertainment plan, family movie plan

Category	Playbooks
Dining	Quick lunch, quick snack, activity before/after movie
Shopping	Category shopping, brand store lookup, luxury shopping plan, last-minute gift, quick errand shopping
Entertainment	Movie showtime lookup, movie night plan, movie + food plan, kid-friendly movie + food
Planning	Date plan, anniversary plan, mall overview, route proximity refinement
Gifts	Gift for girlfriend, gift for family
Budget-conscious	Budget family outing
Solo	Solo visit plan
Combinations	Shopping + dessert combo

Current Limitations

The following are known boundaries of the current version:

Area	Current Behaviour
Language	English only; Arabic is not yet supported
Pricing data	The chatbot cannot confirm live prices — it guides visitors to the right stores
Real-time inventory	Stock availability is not integrated; based on tenant listing only
Promotions	Active promotions are shown when present in the data; not live-synced
Booking / transactions	The chatbot is informational only — it cannot make reservations or process payments
Third-party services	Cannot call taxis, order food delivery, or perform external actions
Image / video	Text-only interface in current form

In Progress (Q2 2026)

These features are actively in development or scoped for the next sprint:

Feature	Description
Arabic language support	Full RTL Arabic-language responses, with cultural tone adaptation for Saudi visitors
Live promotions sync	Real-time pull from the tenant promotion feed — offers reflected within hours of going live
Enhanced personalisation	Returning visitor recognition — remembers stated preferences across sessions
Richer movie data	Showtimes, ratings, trailers linked — not just title and genre
Feedback loop	Explicit thumbs-up/down feedback surfaced to the visitor; fed back into ranking
Admin dashboard	Real-time view of conversation volumes, top queries, and satisfaction signals per mall
Additional malls	Onboarding pipeline validated for 3+ new Cenomi properties

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Future Enhancements (v2.0 Roadmap)

These capabilities are planned but not yet scoped for development:

Feature	Description
Voice interface	Speak your query at a kiosk — response read aloud in Arabic or English
Personalised loyalty integration	Cenomi loyalty card data to personalise recommendations ("You usually shop at Zara — they have a new collection")
Turn-by-turn navigation	In-mall mapping integrated with chatbot responses — "Turn left at Gate 2, then 30m ahead"
Wish list & session save	Visitor can save a list of stores to visit; sent to their phone via QR code
Event-driven promotions	The chatbot proactively surfaces relevant events — "There's a Flash Sale at Centrepoint today"
Multi-modal input	Photo recognition — "What is this store?" from a photo
B2B concierge mode	Corporate and group booking management — team outings, event space inquiries
Sentiment analytics	Real-time dashboard showing sentiment trends across malls — mood-of-the-mall heatmap

Feature	Description
White-label SDK	Package the concierge pipeline as a configurable SDK deployable for any retail/hospitality brand

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Quality Standards

The system is designed to the following principles:

- **LLM-first, rule-free:** All intent classification, context extraction, and response generation is driven by the LLM — no regex, no keyword matching, no hardcoded rules
- **Graceful degradation:** Every error case has a safe fallback — the chatbot never crashes or returns an empty response
- **Privacy-first:** No personally identifiable information is stored; session state is ephemeral and keyed to an anonymous session ID
- **Auditability:** Every response includes a debug envelope capturing the full decision chain — what flow was taken, which playbook matched, what confidence was assigned

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For technical queries, contact the Engineering Team.